How Can Science Support Maritime Decarbonisation?

My journey from shipping to research, policy and implementation

Jeremy Jiajing Chen, Year 4 PhD Candidate
Division of Environment and Sustainability
Hong Kong University of Science and Technology

Today’s lecture has two parts

PART 1

How I ended up in shipping research

30–40 minutes

My shipping background, an unexpected PhD decision, my interest in computers and coding, and the research I do today.

PART 2

How science supports maritime decarbonisation

40 minutes

How evidence helps us measure emissions, compare possible pathways, and move promising ideas towards implementation.

Break between the two parts · Part 2 begins around 11:00

PART 1

How I ended up in shipping research

My journey through family, shipping, computers and one unexpected decision.

Shipping came before research.

My grandfather and father both worked at sea, so shipping had always felt personal to me.

I later studied **International Shipping Management**, followed by **Logistics and Environmental Policy**—first learning how the industry works, and then asking how it could change.

IMAGE PLACEHOLDER

Family connection to shipping or early maritime study

A PhD was not my first choice.

After my Master's, I received an offer for a supply-chain management role at **Huawei**.

I then applied for a short Research Assistant position, mainly to help pay for my flight home.

A ten-minute conversation with Prof. Alexis Lau **(my boss)** changed the direction. The next day, I decided to pursue a PhD.

IMAGE PLACEHOLDER

Huawei career plan, RA opening or the turning-point conversation

I simply enjoyed computer games and coding.

There was no deeper reason at the beginning: I liked spending time on a computer, playing games and writing code.

Research gave me a way to bring these interests into my daily work: working with data, building algorithms and solving difficult problems.

IMAGE PLACEHOLDER

Computer games, coding or an early programming setup

THE FIT

Shipping gave me the subject. Coding and developing algorithms gave me the tools.

Research brought the two together.

Today, I use data to understand how shipping works.

From the shore, we see individual ships.

In the data, we begin to see the whole system: where vessels move, where activity concentrates and where environmental impacts may occur.

IMAGE PLACEHOLDER

A simplified AIS or vessel-activity map showing movement patterns and concentrated activity—not a generic ship photo

The methods are complicated. The logic is simple.

STEP 1

Observe real activity

→

STEP 2

Interpret it with code

→

STEP 3

Turn it into useful evidence

The technical details change. This basic research logic remains.

A paper is not the final destination.

The work matters when evidence enters a real conversation.

It matters even more when that conversation supports a practical next step.

IMAGE PLACEHOLDER

A real photo of Jeremy presenting research to port, policy or industry stakeholders

Research gave me something worth sharing.

With research and data in hand, I began taking the work to different places—and presenting it to different audiences.

PHOTO COLLAGE PLACEHOLDER

Use 3–5 real photos of Jeremy giving presentations or attending academic, policy and maritime-industry events. Mix podium, panel and activity images; add place and year captions later.

The ideal plan is to stay in shipping research.

I want to keep studying how shipping works and how it can change.

Ideally, my research would influence the industry and contribute something useful to the wider world.

IMAGE PLACEHOLDER

A real photo of Jeremy doing maritime research or discussing evidence with shipping-industry partners

FROM MY STORY TO THE RESEARCH

What can science actually do for maritime decarbonisation?

After the break, we will move from my journey
to the practical role of evidence.

Part 2 begins around 11:00

Is shipping a significant source of global carbon emissions?

Before looking at the evidence, which of these views do you agree with?

- Shipping is cleaner than road transport, so its emissions are a lower priority.
- Global trade depends on ships, so strong regulation could carry economic costs.
- A fleet of more than 100,000 vessels is large enough to matter globally.
- We need an emissions estimate before deciding how strongly to regulate shipping.

How can we estimate carbon emissions from shipping?

THE BASIC CALCULATION

Fuel consumption × emission factor = CO_2 emissions

A SIMPLE EXAMPLE

1 tonne of marine diesel × 3.206[1] ≈ 3.2 tonnes of CO_2

The next question is therefore simple: how much fuel does the fleet use?

[1] The exact emission factor depends on the fuel. Around 3.2 is a useful classroom approximation for conventional marine fuel.

But how do we know the fuel use of every ship?

At the start of 2026, the global merchant fleet included around **116,000 vessels**. We cannot collect detailed fuel records from every ship.

ONE SHIP, THEN THE WHOLE FLEET

Maritime big data gives us a moving diary for each ship.

1. See when one ship is sailing, waiting or in port.
2. Combine its activity with ship and engine information to estimate fuel use.
3. Repeat for every observed ship and add the estimates.

From maritime big data to fuel estimates

Movement data reveals how each ship operates over time. The same logic can then be repeated across the fleet.

DIAGRAM PLACEHOLDER

Show one ship leaving a time-ordered movement trail. Label periods of sailing, waiting and port time. Combine this activity with simple ship and engine information to produce one fuel-use estimate. Then zoom out to many ship tracks whose estimates are added into a global total.

A global study provides the evidence

The Fourth International Maritime Organization (IMO) Greenhouse Gas Study estimated that total shipping emitted in 2018:

1,056 million tonnes

of CO_2

2.89%

of global human-caused CO_2 emissions

Science role 1: provide evidence for policy discussion

There is no single decarbonisation pathway

A transition can combine four kinds of action:

1. **Use less energy.** Improve hulls, engines and onboard systems.
2. **Operate ships differently.** Adjust speed, routing, loading and port calls.
3. **Change the energy source.** Use fuels with lower lifecycle greenhouse gas emissions.
4. **Prepare ports.** Provide bunkering, shore power, safety systems and supporting logistics.

The appropriate combination depends on the ship, route, cargo and timescale.

Implementation depends on the wider system

A promising technology still has to pass three practical tests:

1. **Can it work?**
   Are compatible vessels, fuels and port infrastructure available?
2. **Can it pay?**
   Can cargo demand, finance and long-term commitments support the cost?
3. **Can it operate within the rules?**
   Do safety standards and public policy enable implementation?

The transition succeeds when these conditions develop together.

Green Shipping Corridors coordinate action

A Green Shipping Corridor is a route between two or more ports where partners work together to advance zero-emission shipping.

- Start with one route and an existing shipping service.
- Bring cargo owners, ship operators, ports and fuel providers into the same process.
- Align vessels, fuel, infrastructure, finance, standards and policy.

IMAGE PLACEHOLDER

A clean route-focused diagram: Port A ↔ Port B, with ships, cargo and low-GHG fuel partners connected around the route

The idea is attractive. Delivery is difficult.

The 2025 global progress review tracked:

84
active Green Shipping Corridor initiatives

4
had reached the realisation stage

The feasibility wall

Many initiatives remain stalled because zero-emission fuels still cost more than conventional fuels.

Which routes have the strongest starting conditions?

Source: Global Maritime Forum, Annual Progress Report on Green Shipping Corridors 2025.

Maritime big data helps identify promising corridors

The same movement data used to estimate emissions can reveal which routes have the strongest starting conditions.

1. Find routes with regular, repeated movement between ports.
2. Identify the vessels and operators already serving each route.
3. Compare traffic scale and service concentration across candidate routes.

IMAGE PLACEHOLDER

A simplified movement-data map showing many ship tracks, then highlighting one recurring Port A–Port B service; label frequency, traffic scale and operator concentration

Science role 2: identify routes with a credible starting point

Why San Antonio–Hong Kong stands out

2021 ROUTE ANALYSIS · #1 BY ESTIMATED FUEL DEMAND

52
two-way voyages

29
ships serving the route

23.3
average voyage days

243,725
tonnes of estimated annual fuel demand

IMAGE PLACEHOLDER

A Hong Kong–San Antonio route map with recurring voyages shown as ship tracks; add a small Cherry Express or reefer-container cue to show the existing trade lane

Why it matters: regular, long-haul traffic creates **predictable fuel demand** and a meaningful opportunity for cleaner fuels.

Research can inform policy dialogue

April 2025

I presented evidence on the route and its potential as a Green Shipping Corridor.

17 November 2025

San Antonio was announced among Hong Kong’s first Partner Ports.

CONTINUING WORK

Research, stakeholder dialogue and practical coordination continue.

IMAGE PLACEHOLDER

A real April 2025 presentation photo or a public image from the Hong Kong Partner Port announcement

What science contributes

- A more reliable understanding of the emissions problem.
- A transparent basis for comparing possible pathways.
- Evidence on feasibility, uncertainty and trade-offs.
- A shared analytical foundation for coordination.

Policy decisions also reflect institutions, economics, politics and values.

THANK YOU

Questions?

Jeremy Jiajing Chen

Email: jchenhl@connect.ust.hk

Website: https://amoylimee.github.io/

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